Answers

1. **ggplot(mtcars, aes(x = disp, y = mpg)) +**

   geom_point() +

   labs(x = "Engine displacement (cubic inches)",

   y = "Miles per gallon")

We can also fit a linear regression model to the data to quantify this relationship:

lm_fit <- lm(mpg ~ disp, data = mtcars)

summary(lm_fit)

This gives the following output:

Coefficients:

Estimate Std. Error t value Pr(>|t|)

(Intercept) 29.59985 1.22972 24.070 < 2e-16 *

disp -0.04122 0.00471 -8.747 9.38e-10 *

The coefficient for the engine displacement variable (disp) is negative (-0.04122), indicating that as engine displacement increases, mpg decreases. This relationship is statistically significant ($p < 0.001$), suggesting that it is unlikely to have occurred by chance

2. To find out the number of rows and columns in the dataset, we can use the dim() function in R:

library(ggplot2)

data(mpg)

dim(mpg)

This gives the following output:

234 11

3. Categorical variables in mpg include: manufacturer, model, trans (type of transmission), dry (front-wheel drive, rear-wheel, 4wd), fl (fuel type), and class (type of car). Continuous variables in R are called doubles or integers.

'data.frame': 234 obs. of 11 variables:

$ manufacturer: chr "audi" "audi" ...

$ model : chr "a4" "a4" " ...

$ displ : num 1.8 1.8 2 ...

$ year : int 1999 1999 2008 ...

$ cyl : int 4 4 4 ..

$ trans : chr "auto(l5)" "manual(m5)" ...

$ drv : chr "f" "f"...

$ cty : int 18 21 ...

$ hwy : int 29 29 31 ...

$ fl : chr "p" "p" ...

$ class : chr "compact" ...

Based on the above output, we can see that the manufacturer, model, trans, drv, fl, and class variables are all categorical variables

On the other hand, the displ, year, cyl, cty, and hwy variables are continuous variables,

4. ggplot(data = mpg) +

geom_point(mapping = aes(x = displ, y = hwy)) +

facet_wrap(~ class, nrow = 2)

Advantages of using faceting include:

- to compare relationships between variables across multiple subgroups in a more organized and visually appealing way than using the colour aesthetic.
- more flexibility than using the colour aesthetic because we can display many subgroups, and we can also use different types of plots for each subgroup.
- make it easier to identify patterns and trends in the data, particularly if the subgroups have different scales or ranges.

Disadvantages of using faceting include:

- produce many plots, which may be overwhelming or difficult to interpret if there are too many subgroups.
- may not be the best choice if the subgroups are not clearly defined or if there are too few observations in some of the subgroups, as it may not provide enough information to draw meaningful conclusions.

5.

In R, you can use the following geoms to draw different types of charts:

- Line chart: Use `geom_line()` to draw a line chart. This geom connects the points in a scatter plot in the order they appear in the data. If you only want to plot the points without connecting them, you can use `geom_point()` instead.
- Boxplot: Use `geom_boxplot()` to draw a boxplot. This geom displays the distribution of a continuous variable by showing the median, quartiles, and outliers.
- Histogram: Use `geom_histogram()` to draw a histogram. This geom displays the distribution of a continuous variable by dividing it into bins and counting the number of observations in each bin.
- Area chart: Use `geom_area()` to draw an area chart. This geom fills the area under a line chart with color, creating a stacked or cumulative chart.